# Anatomy of a Futonic Mission (from the HEAD down)

*Preprint skeleton — DOCUMENT phase of E-mission-head. Companion to "A First Proof Sprint" (futon6). Format: running example, not monograph — the primary source is `futon6/holes/missions/E-mission-head.md` and its sibling artifacts; this paper reads them rather than repeating them.*

**Status:** draft v2 (2026-06-10, Joe + Fable) — level-0 prose now holds to the plain-language standard (the ARGUE §4.1 discipline applied to the whole paper: boxes may speak the system's native language; the prose around them may not).

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## 1. Two ascents of one mountain

In the software system we work in, the unit of work is called a *mission*: a single document that begins with a short statement of intent (its *HEAD*) and grows, phase by phase, into a finished piece of work. On 10 June 2026 one mission went from a bare statement of intent to a verified design in a single working session — and two different instruments watched it happen.

The first instrument read the mission's **structure**. As each section was written, a live outline showed the document's parts — its phases, sections, and cited methods — appearing as nested, colored blocks, with the phases that had *not* been written yet shown as pale "ghost" lines. The second instrument read the mission's **life**. The same statement of intent was compiled into a small numerical fingerprint, from which the system computed a health report for the newborn work: how confident, what was missing, what would count as failure — together with stated conditions under which the work would declare itself dead.

In René Daumal's *Mount Analogue*, a mountain is climbed by parties on different faces, and what makes the summit real is that the ascents agree. The two readings here are two such ascents: neither reduces to the other, and their agreement — made precise in §5 — is the point.

## 2. The running example

The running example is a mission whose subject is missions themselves: *make the opening statement of intent a typed, checkable object*. Its development is unusually well documented because it documented itself — every phase it grew through was captured by the very tooling it was building. The primary sources (everything below is on disk and quoted from, never paraphrased):

| artifact | reading |
|---|---|
| `E-mission-head.md` (the doc, HEAD→DOCUMENT) | both — the common sourc |
| substrate-2 mission-scope hyperedges (31 scopes) + `*mission-overview*` | scope |
| `E-mission-head.aif.edn` (sigil, vitals, F-nodes, amendments) + `-contra` | AIF |
| `meme.db` `diagram/E-mission-head` (4+3 arrows) + `cascade/E-mission-head-argue` (3+3) | both |
| `mission_head_invariants.clj` (VERIFY V0, 5/5) | AIF (verified) / scope (a s |
| `closure-folds.edn` entries incl. the first `:success false` | the loop's grain |

**♡ vitals (computed from this document)**

confidence 0.33 · xenotype 89% · exotype `00011100`

⊗IF ⊗HOWEVER ✓THEN ✓BECAUSE

*alive but wobbly: xenotype completeness 0.89; strong anchors ['THEN', 'BECAUSE'], weak anchors ['IF', 'HOWEVER'].*

generated 2026-06-10T18:16:29 by `head_exotype_probe.py`

## 3. The scope reading (anatomy)

Read structurally, a mission is a nested system of typed regions of text — *scopes*. The standard life-cycle phases form its spine; sections nest inside phases; records of which methods were chosen, and how they fared, nest inside sections. A detector finds these regions automatically and a panel draws them as colored blocks inside blocks (the same typography used in this PDF). This reading is third-person and anatomical: it says what the mission is *made of*. Its most useful device is the **ghost line**: a life-cycle phase that the standard form expects but the document does not yet contain, drawn in pale outline. Here is the running example's opening statement, exactly as the structural typography renders it:

**HEAD (Joe, 2026-06-10, verbatim sense)**

A mission's HEAD is doing two things at once.
First, in a Jazz or Free Jazz sense, the HEAD is the **theme around which the whole mission is improvised** — ideally in a fully automated way, by following the **patterns of improvisation** that Joe intends to learn from the futon5 work. We are not there yet.
Second, each mission is a **virtual peripheral**, and the HEAD is an **AIF head**: it should be** mappable to the usual AIF terminal vocabulary** — priors, preferences, observations, policies — which would allow us to understand, in typed terms, **what it means to satisfy the requirements of the mission**.
Concretely, the HEAD should therefore carry an extraction of **concepts** — per the **Interest Network** — named entities like "storage", "agents", "patterns", "meme graph", "futon3b", "pattern cascades", obtainable just by *reading the HEAD*. The current kernel extraction is unigram-grade; the upgrade is Interest-Network-grade extraction over the mission corpus, feeding the mission-mode concept lane that already exists. The satisfaction-conditions half connects to E-ground-G: a HEAD mapped to AIF terminal vocabulary is what grounded-G ultimately discharges against.

## 4. The organism reading (physiology)

Read the other way, the mission is a small organism. (In-house this is called the *AIF reading*, after **active inference** — a framework from computational neuroscience in which an organism maintains a model of itself and acts to keep that model's predictions true. Nothing below requires more of the theory than that sentence.) Its statement of intent is rewritten in a four-part design-pattern form (*if / however / then / because*) and compiled into a numerical fingerprint — the system borrows the word *sigil*, after an earlier experiment in which simulated ants were animated by installing exactly such fingerprints. From the sigil the system computes birth vitals; the design states in advance what observations would prove it wrong; the work even argues for its own existence against the strongest objections we could write down, and carries a clause that declares it dead, publicly, if its parts are not put to use by a deadline. This reading is first-person and physiological: it says what the mission *wants* and whether it is *well*. Its most useful device is the **open arrow**: a named gap whose target is specified but whose construction is still owed. The moment the organism first spoke — the fingerprint compilation, from the source:

**3.3.1 v0.1 RESULT — the Golemization run (2026-06-10)**

The probe's pattern-mode ran on §3.3's recast (same projector, section-wise):

| section | bits | conf | nearest anchor | cos |
|---|---|---|---|---|
| IF | `00011100` | 0.28 | hexagram-49 **Ge** (molting/revolution) | 0.284 |
| HOWEVER | `00011001` | 0.36 | hexagram-13 **Tongren** (fellowship) | 0.306 |
| THEN | `10101100` | 0.32 | iiching/exotype-248 | **0.454** |
| BECAUSE | `11001000` | 0.37 | hexagram-31 **Xian** (mutual influence) | **0.461** |

**xenotype-32 `00011100·00011001·10101100·11001000`, mean-conf 0.33** (baseline 0.29). Findings, honestly weighted:

1. **Qualitative unlock confirmed: the sigil now exists.** The xenotype was *uncomputable* before the recast; the Golem has a complete shem.
2. **Bit-confidence lift is modest** (+0.04). Sectioning helps; it does not transform the projection. (Caveat: v0.1 reused the whole-pattern projector per section; the bridge's official path trains per-section projectors — some headroom there.)
3. **The real signal moved into anchor proximity, asymmetrically**: THEN and BECAUSE land at cos $\approx 0.46$ — a third stronger than the whole-text best (0.349) — while IF/HOWEVER stay weak ($\approx 0.29$). Reading: **the actionable half of a HEAD (move + warrant) projects into pattern-space well; the intent/gap half is mission-specific prose the pattern-grain anchors don't cover.** Supports next-car 3: HEAD-grain anchors (missions with known outcomes) for the IF/HOWEVER half.
4. The oracle stayed coherent: intent→*molting*, gap→*fellowship*, warrant→*mutual influence*. Recorded as flavor, not evidence.

## 5. The synthesis (the new thing)

The two readings are not merely parallel descriptions; they are **coupled, the way observation and model are coupled in a living thing**:

> ***The structural reading is the organism's observation space; the organism reading is its model of itself. A mission is well-formed exactly when the two cohere, and the coherence is computable.***

The mapping is exact, not poetic:

| structural object | organism object | the coupling |
|---|---|---|
| statement of intent (HEAD) | the model's starting beliefs | the fingerprint compiler |
| the standard phases | stages of a plan | the standard form is an *expectation abou* |
| **ghost line** | **prediction error** | the standard form expects a phase; the d |
| a phase satisfied in passing | a prediction met by an unexpected route | belief updated without the expected obse |
| open arrow | an unmet commitment the model still expects | the deadline clause prices its persistence |
| method chosen / how it fared | action selected / outcome observed | the records make the learning loop legibl |
| stated failure condition | what would *count* as error | written before the construction, never aft |
| the vitals card | interoception (the body sensing itself) | computed *from* the structurally-readable |
| the outline panel | exteroception (the body as seen) | the same document, drawn |

Three consequences worth the paper:

1. **The body-schema claim.** Late in the session, three pieces of wiring were added: the health report is computed from the document, displayed in the outline panel, and fed into the mission's model of itself. That is not plumbing; it is the binding of inner sense to outer sense — the point at which the mission acquires a body schema. Before it, two instruments; after it, one organism with two senses.
2. **Coherence is checkable.** Disagreements between the readings are not embarrassments but findings, surfaced mechanically: structure the model has no use for (decoration); a want with no corresponding structure (a desire nothing can observe — the session caught a cited method that had never actually been written down this way); a ghost the model says is already satisfied (one phase was closed in passing, inside another section, and the detector learned to see it). The session produced live instances of all three.
3. **The synthesis pays one of its own debts.** When the mission argued for its existence, its argument leaned on a design principle that turned out never to have been written into the method library: *two projections of one quantity.* The synthesis above is precisely that principle's missing construction — the structural reading and the organism reading are two projections of one underlying object, stored three ways. Publishing the argument writes the missing principle down.

And the Daumal closure, which the stack supplies on its own: the points where two independent ascents *agree* are the only points one may call real — and the stack already has a name and a discipline for externally-checkable value found on the mountain: **the peradam**. The synthesis predicts where peradams form: at reading-agreements. "One descends, one sees no longer, but one has seen."

And the argument in the register anyone can read — the mission's own plain-language statement, boxed as it appears in the source:

### 4.1 Plain-language argument (the version anyone can read)

When we start a piece of work, we write a short statement of what it is for. Right now that statement is just words: nothing else in the system can read it, check it, or use it. So nobody can tell, at the start, whether a piece of work rests on solid ground or shaky ground — and when work quietly stalls, nothing notices. This has already happened here: a major piece of work was "finished" on paper three months ago, and nobody saw that it was never actually connected to anything.

The proposal: when work begins, turn its opening statement into a small summary the system can read — what the work wants, what is in the way, what it will do, and why. From that, the system can show, from day one, how strong or weak the work looks, what is missing, and what would count as finishing. As the work proceeds, the same readout shows whether it is getting healthier.

What we checked before believing this: we tried it on this very piece of work. It partly worked — the "what to do" and "why" parts came through clearly; the "what we want" part needs better reference material, which we now know how to gather. We also put up the strongest objections we could find, and answered each one:

1. *"These summaries will be built and then ignored, like last time."* Last time, nothing showed that the finished work was never put to use. This time there is a deadline: if the pieces are not in use by then, the work is marked as failed, in a place everyone can see.
2. *"People will write their statements to score well."* The readout is never used as a target or a gate. It is information only, checked occasionally and independently.
3. *"This will make creative work bureaucratic."* The summary describes only the starting point, never the path. The path stays free — that freedom is the point of working this way, and it is kept on purpose.

Every claim above comes with a stated test that would prove it wrong, and the first of those tests has already been run and passed.

Why this is needed, in one line: **work should be able to say what it is for, in a form that lets anyone — or anything — check how it is doing.**

## 6. Method appendix (planned)

How the session was actually run: a human operator and several AI agents sharing one learning loop — every method-use recorded with its outcome (including the record-keeping's first honestly-logged *failure*), build work handed to worker agents with acceptance criteria and reviewed on return, and each claim checked against the system's stores before being written here. Technical cross-references for in-house readers are kept in the source repositories rather than in this text.

## 7. Relation to "A First Proof Sprint"

"A First Proof Sprint" follows a single mathematical proof finding its way through this same system. The present account sits one level up: it follows the *unit of work itself* — the mission — through the same passage, the container examining the concept of containment. Same house, different floor.

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*TODO (next sessions): §3/§4 prose passes over the primary source; figures (the panel at frames 1→6; the bout's cross-edge diagram; the vitals card); §5 worked dis/agreement instances with store queries; decide venue + whether the two-projections flexiarg ships with the paper or before it.*